

Math - home assignment

Game Setup

Grid: 5 reels × 3 rows, 20 fixed paylines.

Symbols and reel weights (each reel has its own strip of 30 positions; the weight shows how many times the symbol appears on that reel's strip):

Symbol	Reel 1	Reel 2	Reel 3	Reel 4	Reel 5
Wild	1	1	2	1	1
Symbol 1	2	2	2	2	3
Symbol 2	3	3	3	3	3
Symbol 3	4	4	4	5	4
Symbol 4	5	5	5	5	5
Symbol 5	6	6	6	6	6
Symbol 6	9	9	8	8	8

(Each column sums to 30 - the length of the reel strip.)

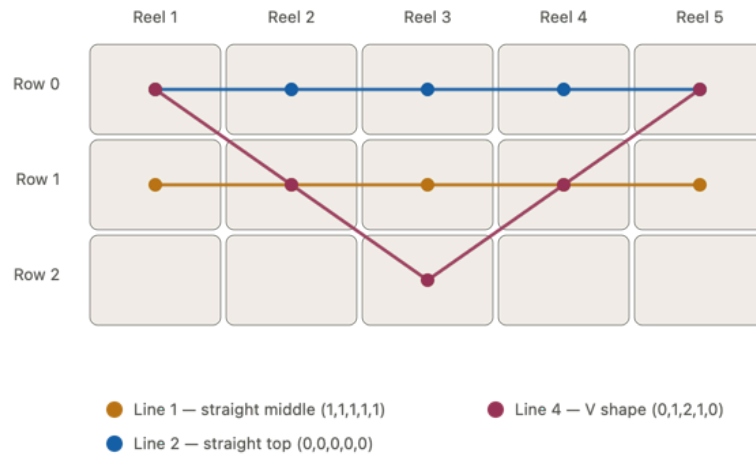
Paytable (payout as a multiplier of total bet, for a matching run of symbols left-to-right on an active line):

Symbol	3 of a kind	4 of a kind	5 of a kind
Wild	5x	20x	100x
Symbol 1	4x	15x	60x
Symbol 2	3x	10x	40x
Symbol 3	1x	5x	15x
Symbol 4	0.8x	4x	12x
Symbol 5	0.5x	2x	8x
Symbol 6	0.4x	1.5x	6x

Wild rule: Wild substitutes for all symbols to form a winning combination.

If multiple winning combinations are possible on a single payline (e.g., due to Wilds), the game pays only the highest value combination for that line.

Paylines: The grid has 3 rows, indexed 0 (top), 1 (middle), 2 (bottom). The 20 active paylines are defined below as the row index on each of the 5 reels, left to right starting strictly from the leftmost reel (Reel 1):



Line #	Reel 1	Reel 2	Reel 3	Reel 4	Reel 5
1	1	1	1	1	1
2	0	0	0	0	0
3	2	2	2	2	2
4	0	1	2	1	0
5	2	1	0	1	2
6	0	0	1	2	2
7	2	2	1	0	0
8	1	0	0	0	1
9	1	2	2	2	1
10	0	1	1	1	0

11	2	1	1	1	2
12	1	0	1	2	1
13	1	2	1	0	1
14	0	1	0	1	0
15	2	1	2	1	2
16	1	1	0	1	1
17	1	1	2	1	1
18	0	0	2	0	0
19	2	2	0	2	0
20	0	2	0	2	0

Tasks

1. Analytical calculation

Attempt to calculate theoretical RTP and hit frequency using a direct probability formula. If you judge the analytical approach impractical for 20 lines with wild substitution and explain in writing why.

2. Simulation

Write a simulation that:

- Generates random spins according to each reel strip's weights
- Evaluates winnings across all 20 lines for every spin
- Runs a minimum of 1,000,000 spins
- Outputs: theoretical RTP (%), hit frequency (%), win size distribution (histogram or table by bins: 0x, 0-1x, 1-5x, 5-20x, 20-50x, 50x+)

3. Short written commentary

- List explicitly all assumptions you made about the game model that are not stated in this document.

- What is the volatility profile of this configuration based on your simulation results (low/medium/high), and why?
- If you needed to increase volatility while keeping RTP roughly the same, which 2-3 parameters would you change first, and why those specifically?
- Suppose the target RTP for this game is 96%. What would be your practical process for bringing the configuration to that target quickly, without rewriting everything from scratch